

Explainability is Relational: A Receiver-Dependent Framework for AI Interpretability

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https://github.com/RedaAZIZI/impact_social

Abstract

Explainability in AI is treated as an intrinsic property of the model. We argue that this is a category error: explainability is a property of the *relation* between a model M and a receiver R , each endowed with its own representational basis. Classical XAI (feature attributions, surrogate trees) is the degenerate case in which the bases of M and R are assumed identical. We propose (i) a formal definition: M is explainable for R if and only if there exists a decision graph, expressed in R 's predicate vocabulary and of bounded complexity, that simulates M to fidelity ϵ ; (ii) a contrastive, composable *why-operator* W that turns explanation into an interactive protocol whose cost (number of questions) measures explainability; (iii) controlled experiments on a synthetic world whose explanatory ground truth is known. Results: the cost of explainability grows monotonically with the misalignment angle between bases (fidelity AUC $0.918 \rightarrow 0.745$); reconstructing M 's graph through successive why-questions converges in ~ 6 questions under an aligned basis (fidelity 97.9%) and fails to converge under a misaligned one (plateau at 71%) — same model, same answers: only the receiver's basis changes. On real data, the framework predicts where natural misalignment appears: absent on a credit-scoring task whose expert concepts do not carry the signal (a clean, pre-registered negative result), significant on a predictive-maintenance benchmark whose failure laws are written in compound physical quantities ($+0.046$, $p < 10^{-3}$) — and modulated by the internal basis of the model itself. Finally, under continuous drift, a graph maintained through dialogue (one sentence per drift, zero labels) outperforms a network retrained on 75,000 examples, with no cumulative forgetting: we conjecture learning by graph editing as a candidate post-gradient paradigm.

1 Introduction: the hidden assumption of XAI

Between humans, explanation works because a common ground — language, concepts, culture — is shared by default. That channel was never formalized because it never failed. With AI systems the assumption collapses: the internal features of a neural network do not coincide with human concepts. Yet the standard explainability toolbox — SHAP [1], LIME [2], saliency maps, surrogate trees [3, 4] — implicitly assumes that the axes of the model *are* the concepts of the receiver. These methods compute an explanation in the basis of M and deliver it, unchanged, to R .

Thesis. *Explainability is not a property of M . It is a property of the triple (M, R, Q) : the model, the receiver's representational basis, and the contextual question being asked. Classical XAI is the special case $\text{base}(M) = \text{base}(R)$ with $Q = \text{"why this output?"}$ and no explicit contrast.*

This paper is a position paper with preliminary evidence.

Contributions.

- (i) **A formal framework** in which explainability is defined relative to a receiver R and a contextual question Q , with two operational measures — the explainability curve and the number of why-queries — and classical XAI recovered as the aligned degenerate case (Section 2).
- (ii) **Controlled evidence.** On a synthetic world whose explanatory ground truth is known by construction, the cost of explanation is governed by the alignment between the bases of M and R , not by M alone: it grows monotonically with basis misalignment (Section 3.1), is invariant to monotone reparametrizations (Section 3.2), reappears identically in an interactive protocol (Section 3.3), and depends asymmetrically on the question (Section 3.4).
- (iii) **The relation meets reality.** On real data the framework predicts *where* natural misalignment appears; we confirm both the negative and the positive prediction with pre-registered criteria, and find the effect modulated by the model's own internal basis (Section 4).

- (iv) **The inverse channel.** Because explanation is a channel, it can be used in reverse: editing an explainable graph through dialogue outperforms gradient-based adaptation under drift, with zero cumulative forgetting by construction; we state the corresponding conjecture explicitly (Section 5).

If explanation is an interactive protocol — a series of “why?” questions — then the same channel serves learning: the inverse of the why-operator inserts graph fragments supplied by R into M ’s explainable graph. The long-term perspectives are (a) AI that learns in dialogue by graph editing — local, instantaneous, traceable — rather than by gradient descent; (b) contextual compression: a large model distilled on the fly, per context, into a small, locally executable, fully explainable graph; (c) personal AI that grows with its user. The performance–interpretability trade-off [5] is real but is usually evaluated on static benchmarks; in an interactive regime, a graph correctable in one sentence can beat a frozen network that requires retraining. This is a falsifiable thesis, and Sections 3–5 are its first controlled test.

2 Formal framework

2.1 Definitions

Definition 1 (Receiver, candidate explanation). A *receiver* is a pair $R = (V_R, C_R)$: a vocabulary of predicates V_R — the tests R can evaluate and understand — and a capacity constraint C_R bounding the size of the structures R can hold. A *candidate explanation* of a model M for R is a decision graph G whose internal nodes are predicates of V_R , whose edges carry the conditional flow, and whose leaves are outputs.

Definition 2 (Relational explainability). Let $\text{fid}(G, M)$ be the agreement between G and M on the data distribution. The explainability of M for R at fidelity level ε is the size of the smallest faithful explanation,

$$\text{Expl}(M, R, \varepsilon) = \min\{|G| : G \text{ expressed in } V_R, \text{fid}(G, M) \geq 1 - \varepsilon\}, \quad (1)$$

and M is *explainable for R* at level (ε, K) if and only if $\text{Expl}(M, R, \varepsilon) \leq K$. The *explainability curve* of the pair (M, R) maps each budget $|G|$ to the maximal fidelity attainable within it — the explainability signature of M relative to R ; its area under the curve (AUC) is the associated scalar metric.

Degenerate case. If V_R consists of threshold predicates on M ’s own input features (the identity basis), Equation (1) reduces to classical tree distillation: standard XAI is relational explainability restricted to perfect alignment.

Two axes of complexity interact in Definition 2: structural complexity (the size of the graph) and predicate complexity (the vocabulary V_R). A small graph with dense predicates ($w \cdot x > \theta$) is not readable; an axis-aligned graph is. The “receiver’s basis” is V_R .

2.2 The contextual question Q

Following van Fraassen’s pragmatic theory of explanation [6], every why-question is contrastive (“why A rather than B ?”) and context selects the contrast; see also Miller [7]. In our framework, the answer to Q is the minimal subgraph where the paths toward A and toward B diverge (the pivotal nodes). Counterfactual explanations [8] are a special case. Full explainability then becomes an expectation (or worst case) over the distribution of questions Q of the usage context.

2.3 The why-operator W

$W(x, \text{contrast}) \rightarrow$ minimal pivotal subgraph, expressed in V_R . Three required properties: **minimality** (no superfluous node), **translatability** (output in the receiver’s vocabulary), **composability** (answers assemble). Explanation becomes a protocol: R poses successive W -queries until sufficient simulatability [9, 10] is reached — the “ah, I see” formalized. This yields a second measure of explainability: *the number of W -questions needed to reconstruct M to fidelity ε* . The theoretical anchor is Angluin’s exact learning from queries [11] — W is a richer query than membership or equivalence. The connection to interactive proofs and debate [12], with the receiver as a bounded verifier, is left as a one-line perspective.

2.4 Related work

Representational alignment is measured by CKA [13], Procrustes analysis, and model stitching [14]; we use controlled rotations to *manipulate* alignment rather than measure it. Simulatability as the operational test of explanation quality

Table 1: Experiment 1 — fidelity to M as a function of leaf budget and receiver misalignment θ (10 seeds, mean $\pm$ std). Same model, same information (rotations are bijections); only the receiver’s basis varies.

Budget	$\theta=0$	$\theta=0.2$	$\theta=0.4$	$\theta=0.8$	$\theta=1.5$
2	0.696 $\pm$ 0.015	0.678 $\pm$ 0.013	0.654 $\pm$ 0.014	0.595 $\pm$ 0.030	0.549 $\pm$ 0.041
4	0.834 $\pm$ 0.011	0.794 $\pm$ 0.026	0.737 $\pm$ 0.035	0.678 $\pm$ 0.030	0.641 $\pm$ 0.035
8	0.936 $\pm$ 0.009	0.895 $\pm$ 0.014	0.835 $\pm$ 0.014	0.736 $\pm$ 0.017	0.705 $\pm$ 0.023
16	0.976 $\pm$ 0.007	0.934 $\pm$ 0.012	0.879 $\pm$ 0.014	0.791 $\pm$ 0.018	0.752 $\pm$ 0.020
32	0.977 $\pm$ 0.007	0.944 $\pm$ 0.010	0.904 $\pm$ 0.015	0.829 $\pm$ 0.018	0.795 $\pm$ 0.016
64	0.975 $\pm$ 0.008	0.949 $\pm$ 0.007	0.917 $\pm$ 0.013	0.852 $\pm$ 0.014	0.824 $\pm$ 0.019
128	0.975 $\pm$ 0.008	0.952 $\pm$ 0.008	0.920 $\pm$ 0.012	0.872 $\pm$ 0.014	0.841 $\pm$ 0.017
256	0.975 $\pm$ 0.008	0.950 $\pm$ 0.008	0.921 $\pm$ 0.010	0.878 $\pm$ 0.013	0.855 $\pm$ 0.014
AUC	0.918	0.887	0.846	0.779	0.745

goes back to Doshi-Velez and Kim [9] and Hase and Bansal [10]. Tree distillation of neural networks (TREPAN [3], soft decision trees [4]) is, in our terms, the aligned degenerate case. Rudin [5] argues for interpretable models in high-stakes settings; our framework makes her trade-off receiver-dependent. Catastrophic forgetting [15, 16] and its mitigations such as EWC [17] motivate Section 5: local graph edits do not mitigate the problem, they remove it by construction. Model editing in LLMs (ROME [18], MEMIT [19]) edits opaque weights through fragile localization heuristics; here we edit an explainable structure through the very channel of explanation — editing is a by-product of the framework, not an ad hoc technique. Machine teaching [20] anticipates the data-efficiency of a semantic teaching channel. Finally, a ReLU network is exactly an exponential-size decision graph, so the question becomes one of compressibility — a Kolmogorov-style analogy restricted to graphs.

3 Controlled experiments: the receiver in the equation

All experiments in this section use a synthetic world where the explanatory ground truth is known by construction — the only setting in which the receiver’s basis can be manipulated exactly. The world has 6,000 objects with 5 interpretable attributes $\sim U[0, 1]$ (size, hue, shape, position x , position y) and a hand-written ground-truth rule graph G^* with 5 branches (e.g. *if size > 0.6 and hue > 0.5 $\rightarrow$ class A*). The opaque model M is an MLP (2×50 units) trained on $(X, G^*(X))$, reaching accuracy 0.9762 ± 0.0059 (10 seeds). Code, seeds, and figures are in the repository (Section 7).

3.1 Experiment 1: the cost of explanation grows with basis misalignment

Protocol. Receivers are defined by their basis: a rotation of strength θ applied to the 5 dimensions ($R_\theta = \exp(\theta A)$ with A a normalized random antisymmetric matrix; $\theta = 0$ is the identity), $\theta \in \{0, 0.2, 0.4, 0.8, 1.5\}$. Explainability is measured by distilling M into trees with threshold predicates *in the receiver’s basis*, for leaf budgets 2 to 256; fidelity is agreement with M on held-out data; 10 seeds, mean $\pm$ std. The falsification criterion was set a priori: superposed curves would void the concept.

Results (Table 1, Figure 1). The explainability AUC decreases strictly monotonically with θ : 0.918 at $\theta=0$ down to 0.745 at $\theta=1.5$. The minimal budget for fidelity ≥ 0.95 is 16 leaves at $\theta=0$, 128 at $\theta=0.2$, and is never reached (>256) for $\theta \geq 0.4$.

Interpretation. Same model M (not one weight changes), same information (rotations are bijections). Only the receiver’s basis varies, and the explainability curves diverge monotonically with θ . Explainability is therefore not a property of M alone.

3.2 Experiment 1b: what costs is mixing, not reparametrization

Applying a coordinate-wise monotone deformation on top of the rotation ($v = \tanh(s \cdot (x - 0.5)) \cdot R_{\theta=1.5}$), $s \in \{2, 6\}$, 5 seeds) leaves the AUC unchanged: 0.7432 vs. 0.7433 vs. 0.7433 for the linear, $s=2$, and $s=6$ cases.

Proposition 1 (Monotone invariance; empirical). *For a receiver whose vocabulary consists of threshold predicates, the cost of explainability is invariant under coordinate-wise monotone deformations of its basis.*

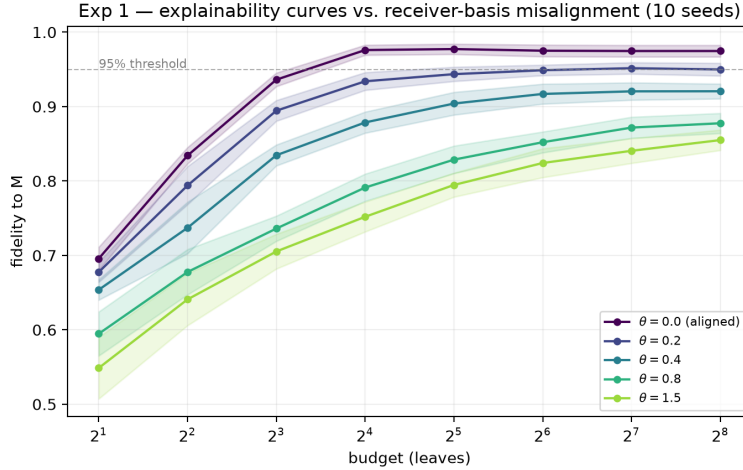


Figure 1: Experiment 1 — explainability curves (fidelity vs. budget, bands $\pm$ std over 10 seeds) for receivers at increasing basis misalignment θ . Not a single weight of M changes across curves.

Table 2: Experiment 2 — fidelity to M of the reconstructed model vs. number of why-questions (5 seeds). Same M , same answers to every query; only the receiver’s basis differs.

Questions	R aligned	R rotated ($\theta=1.5$)
0	0.457 $\pm$ 0.010	0.457 $\pm$ 0.010
3	0.727 $\pm$ 0.089	0.625 $\pm$ 0.046
6	0.921$\pm$0.022	0.688 $\pm$ 0.024
9	0.966 $\pm$ 0.002	0.695 $\pm$ 0.021
12	0.977 $\pm$ 0.006	0.703 $\pm$ 0.027
15	0.979$\pm$0.007	0.713$\pm$0.024

The intuition is immediate — a threshold transports through any monotone function, so threshold trees depend only on coordinate orders — but we state Proposition 1 as an empirical claim: it was confirmed on every test bed of this paper, synthetic and real, 11 runs out of 11, often at exactly 0.0000 (Sections 4.1 and 4.2); the formal proof is left to future work. Its consequence sharpens the concept: what costs is the *mixing* of dimensions, not their reparametrization — the relevant misalignment is structural, not metric.

3.3 Experiment 2: explanation as a measurable protocol

Protocol. Same world, same M ; the reference graph of M is a 32-leaf tree distilled in the original basis. The operator $W(x)$ answers “why is x classified c ?” with the decision path of x (a conjunction of threshold predicates) plus its class — minimality via the path, implicit contrast versus the other leaves. Dialogue protocol (active learning): the receiver starts from a trivial majority-class predictor; at each turn it picks a point where it disagrees with M , asks W , integrates the rule, and repeats (15 questions max, 5 seeds). The *aligned* receiver stores the rule verbatim; the *rotated* receiver ($\theta=1.5$) perceives the world only in its own basis, so each received rule — an axis-aligned region for M , an oblique polytope for it — must be approximated by a small 8-leaf tree in *its* vocabulary.

Results (Table 2, Figure 2). The aligned receiver reaches 90% fidelity in 6 questions and 97.9% at 15; the rotated receiver never reaches 90% and plateaus at $\sim 71\%$. Structurally, G^* has ~ 7 branches and ~ 6 questions suffice: the dialogue reconstructs the structure at roughly one rule per question.

Interpretation. Explainability-as-protocol is measurable: the number of why-questions is small and finite when bases are aligned, and the dialogue fails to converge when they are misaligned — *with identical answers from M* . Two independent protocols (static distillation, interactive dialogue) confirm the same thesis.

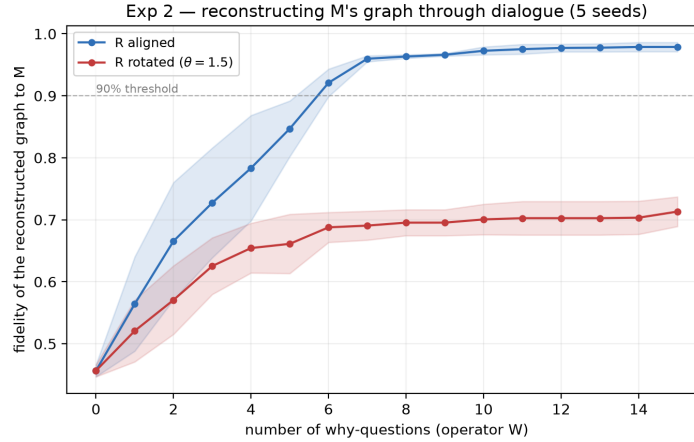


Figure 2: Experiment 2 — reconstruction of M by successive why-questions. The aligned receiver reaches 90% fidelity in 6 questions; the misaligned receiver plateaus near 71% with the same answers from M .

Table 3: Experiment 5 (German Credit) — expert vs. raw vocabulary, $n = 10$ seeds, paired per-seed statistics. The pre-registered falsification criterion fired: a clean negative result.

Run	AUC raw	AUC expert	Paired gap	Paired t	Wilcoxon
v0 (MLP)	0.733 ± 0.025	0.727 ± 0.019	-0.006 ± 0.018	$p = 0.32$	$p = 0.48$
v0.1 (GBT + queries)	0.803 ± 0.026	0.801 ± 0.021	-0.002 ± 0.019	$p = 0.72$	$p = 0.83$

3.4 Experiment 4: explainability depends on the question

On the reference graph of M , we measure the size of the contrastive answer to “why a rather than b ?” — the minimal number of diverging predicates between the instance’s path and the closest path leading to b , weighted by leaf coverage. The answer size ranges from 2.95 predicates (B rather than A) to 6.39 (B rather than C), and the relation is asymmetric: A-vs-B costs 5.81 while B-vs-A costs 2.95. The same model is twice as “explainable” for some questions as for others, consistently with the pragmatic theory: explanation is not symmetric in the contrast. Full explainability should thus be defined as an expectation over the question distribution of the usage context, $\text{Expl}(M, R, \bar{Q}) = \mathbb{E}_{\bar{Q}}[\text{answer size}]$.

4 Contact with real data: a predictive diptych

If misalignment is real, it should occur *naturally*: some domains describe the world in compound expert concepts, others do not. The framework makes a falsifiable prediction about where an expert vocabulary should dominate a raw one — and we tested it on two public datasets with pre-registered hypotheses and falsification criteria, reporting the negative result with the same prominence as the positive one. Both experiments use $n = 10$ seeds and per-seed paired statistics (paired t -test and Wilcoxon signed-rank), and both include the monotone control of Section 3.2 (log/power transforms of raw features), predicted inert.

4.1 Experiment 5 (negative): German Credit

Protocol. Statlog German Credit [21]: 1,000 applications, 20 attributes, binary good/bad outcome (70/30). Receiver vocabularies: *raw* (features as-is); *expert* = raw + compound credit concepts (installment = amount/duration, burden = installment $\times$ repayment rate, exposure = amount $\times$ number of credits); monotone control. Registered protocol v0 used an MLP (2×50); a documented amendment v0.1 switched to gradient boosting plus 20,000 extra distillation queries after the v0 MLP (accuracy 0.72 against a 0.70 majority class) turned out to have almost no structure to explain — the methodological lesson, incorporated into all subsequent protocols, is to normalize the explainability curve by the fidelity of the trivial model. Falsification criterion set a priori: an expert–raw AUC gap below one inter-seed standard deviation means no natural misalignment on these data.

Table 4: Experiment 6 (AI4I 2020) — engineer vs. raw vocabulary, $n = 10$ seeds, paired per-seed statistics, class-balanced fidelity. The gap depends on the internal basis of M itself.

M	AUC raw	AUC engineer	Paired gap	Paired t	Wilcoxon
MLP	0.702±0.050	0.748±0.048	+0.046±0.026	$t = 5.37, p = 4.5 \times 10^{-4}$	$p = 0.002$
GBT	0.760±0.022	0.755±0.017	−0.005±0.027	$p = 0.56$	$p = 0.56$

Results (Table 3). No dominance of the expert vocabulary, in either run. The monotone control is inert as predicted (-0.0008 ± 0.0016 in v0; 0.0000 in v0.1) — the invariance of Section 3.2 holds on real data.

Interpretation. The falsification criterion spoke: no natural misalignment on German Credit. Two readings: (a) the predictive signal of this dataset lives in categorical variables (account status, credit history) shared by both vocabularies, and the central expert concept of credit — the debt-to-income ratio — is not computable from the available attributes (no income); (b) more deeply, expert-vocabulary dominance requires the world’s generating process to be itself written in compound quantities. That is precisely what Experiment 6 tests.

4.2 Experiment 6 (positive): predictive maintenance

Protocol. AI4I 2020 Predictive Maintenance [22]: 10,000 points, 5 sensors (air and process temperatures, rotational speed, torque, tool wear). The benchmark’s failure modes are *written* in compound quantities: heat dissipation (ΔT and speed), power (torque $\times$ angular speed), overstrain (wear $\times$ torque). Hypothesis registered a priori: on physical ground — where the generating process is written in compound quantities — the engineer’s vocabulary (sensors + ΔT , power, strain) dominates the raw-sensor vocabulary; same falsification criterion as Experiment 5. Fidelity is class-balanced (failures are rare — the lesson of Experiment 5), budgets 2 to 256 leaves, $n = 10$ seeds, paired tests, and two opaque models: an MLP (2×50) and gradient boosting.

Results (Table 4). For the MLP, the engineer’s vocabulary dominates ($+0.046 \pm 0.026$, paired $t = 5.37, p = 4.5 \times 10^{-4}$, Wilcoxon $p = 0.002$). For gradient boosting, the gap vanishes ($-0.005 \pm 0.027, p = 0.56$). Monotone controls are inert in both cases (Proposition 1). Balanced accuracy of M is 0.62 (MLP) and 0.71 (GBT); the residual failure modes of AI4I are known to be nearly unlearnable.

Interpretation. To our knowledge, this is the first controlled demonstration of *natural* basis misalignment on real data. Together with Experiment 5 it forms a diptych: no dominance where the world is not written in compound concepts (credit), clear dominance where it is (physics) — natural misalignment exists, and the framework predicts where. The GBT result is the structural surprise: a tree ensemble is built from axis-aligned splits on raw features — *its internal basis is the raw vocabulary by construction* — and the gap disappears; the MLP mixes dimensions the way the physics does, and the gap appears. Same data, same receivers, different M , different explainability relation: the relational thesis observed *on the model side*, everything else held fixed. Caveats: one dataset per verdict; expert vocabularies chosen by the authors, not learned; rare classes limit the learnable structure of M .

5 The interactive regime: the channel that explains is the channel that learns

The why-operator has an inverse: where W extracts a rule from M ’s explainable graph, W^{-1} inserts or edits one, supplied by the receiver in the shared vocabulary. Because an edit touches exactly the rules it names, regions outside the edit are intact *by construction* — catastrophic forgetting [15, 16] is not mitigated, it is removed. This section tests the resulting learning regime, first on a single correction, then under continuous drift.

5.1 Experiment 3: one sentence vs. thousands of examples

Protocol (concept drift). The world changes: the rule “shape $> 0.7 \rightarrow B$ ” of G^* becomes “shape $> 0.7 \rightarrow A$ ”. M , trained on the old world, drops to 0.741 accuracy. Route A (our framework): the explainable graph of M (rule list extracted from the 32-leaf reference tree) is corrected by *one* sentence of dialogue — “when shape > 0.7 , it is A now” — implemented as W^{-1} editing the class of the concerned rules; cost: one interaction, zero gradients. Route B (standard paradigm): the MLP is fine-tuned (warm start, 50 epochs) on $n \in \{10, \dots, 3000\}$ freshly labeled examples. Accuracy on the new world, 5 seeds.

Table 5: Experiment 3 — adaptation to a rule flip: one graph edit through dialogue vs. fine-tuning on labeled examples (5 seeds).

Method	Cost	Accuracy (new world)
M without adaptation	—	0.741 ± 0.012
Graph edit (dialogue)	1 sentence	0.990 ± 0.004
Fine-tuning	10 examples	0.758 ± 0.046
Fine-tuning	100 examples	0.893 ± 0.012
Fine-tuning	1000 examples	0.947 ± 0.004
Fine-tuning	3000 examples	0.968 ± 0.004

Table 6: Experiment 7 — mean accuracy over 15 drifts (5 seeds). The dialogue-maintained graph uses 15 sentences and zero labels; all four pre-registered kill conditions survived.

Method	Total supervision (15 drifts)	Mean accuracy (drifts 1–15)
Frozen MLP	—	0.624 ± 0.002
Fine-tuned MLP	1,500 labels	0.854 ± 0.018 (forgetting sawtooth, min 0.740)
Retrained-from-scratch MLP	75,000 labels	0.981 ± 0.002
Graph + dialogue	15 sentences, 0 labels	0.990 ± 0.003
Graph, noisy sentences ± 0.05	15 sentences	0.984 ± 0.008
Graph, noisy sentences ± 0.10	15 sentences	0.971 ± 0.018
Graph, misaligned basis $\theta=1.5$	15 sentences	0.801 ± 0.028

Results (Table 5). One sentence of dialogue outperforms 3,000 examples of fine-tuning (0.990 vs. 0.968). The edit is instantaneous, local, and traceable.

Methodological fair play. The comparison is deliberately *not* information-equal: one semantic sentence compresses what thousands of examples express — that is the thesis, not a confound. The explanation channel (W) and the learning channel (W^{-1}) are the same channel used in both directions. Note also that Route A requires the correction to be expressible in the shared vocabulary: under a misaligned basis (Experiments 1–2) this channel degrades, which ties the experiments into a single narrative — and Experiment 7 quantifies exactly that tax.

5.2 Experiment 7: a graph maintained by dialogue under continuous drift

Protocol (pre-registered; kill conditions published before the run — the git record is the evidence). The reference world G^* undergoes 15 successive drifts: at each step the class of one branch flips (deterministic cyclic schedule). Competitors: (a) the explainable graph (32 leaves distilled from the initial M) maintained by dialogue — *one* sentence per drift (“in the region (branch condition), the class is now c ”), interpreted against the receiver’s own unlabeled experience (any rule whose firing region lies mostly inside the sentence’s region is re-classified); zero labeled examples; (b) the frozen MLP; (c) the MLP fine-tuned on 100 fresh labeled examples per drift (50 epochs); (d) the MLP retrained from scratch on 5,000 examples per drift. 5 seeds. Stress variants: noisy sentences (thresholds perturbed by ± 0.05 and ± 0.10 — human imprecision); receiver in a misaligned basis ($\theta = 1.5$). Four kill conditions were set a priori: K1, the graph falls below fine-tuning over the last 5 drifts; K2, accuracy of a never-edited region degrades after 15 edits; K3, the advantage over the frozen model vanishes under ± 0.10 noise; K4, the misaligned graph falls below the frozen model.

Results (Table 6, Figure 3). Fifteen sentences with zero labels outperform 75,000 labels of from-scratch retraining (0.990 ± 0.003 vs. 0.981 ± 0.002). All four kill conditions survived. Integrity of the never-edited region after 15 edits: $0.992 \rightarrow 0.994$ — zero cumulative degradation (K2: locality holds over 15 edits). Human imprecision of ± 0.10 costs only ~ 2 points (0.971). The misaligned receiver pays a measured *expressibility tax* of ~ 0.19 (0.801), quantitatively linking Experiments 1–2 to Experiment 3. An unplanned control: the cyclic schedule returns the world to its initial state at drift 12, where the frozen model climbs back exactly to its original level — the drift mechanics are sound.

Conjecture 1 (Post-gradient learning). *Learning by graph editing in dialogue is a candidate paradigm for post-gradient learning: data-efficient because the channel is semantic, and continual by construction because edits are local.*

Conjecture 1 attacks head-on the two best-documented weaknesses of deep learning: data inefficiency (one sentence compresses thousands of examples, Experiment 3) and catastrophic forgetting (locality removes the problem rather than mitigating it, Experiment 7, condition K2). We state it as a conjecture deliberately: it is supported by controlled synthetic evidence, not an established claim.

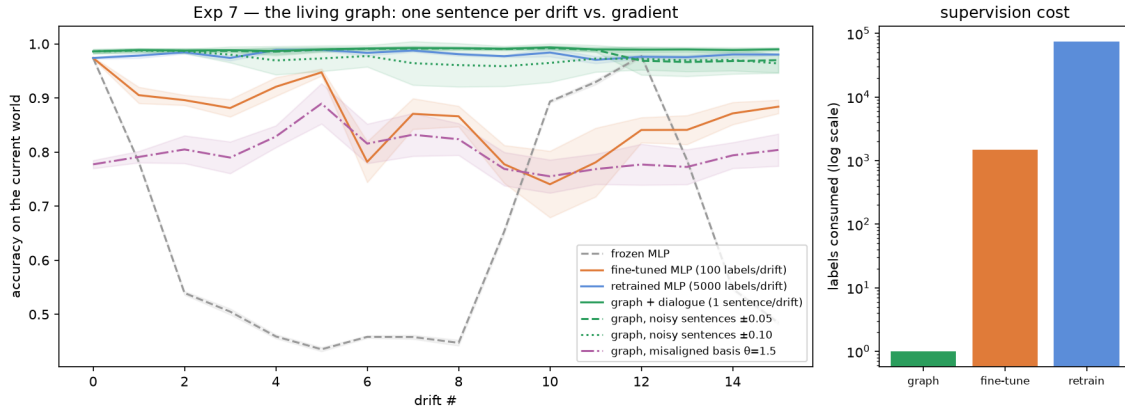


Figure 3: Experiment 7 — accuracy along 15 successive drifts for the four competitors and the stress variants. The dialogue-maintained graph (15 sentences, zero labels) dominates throughout; the never-edited region shows zero cumulative degradation.

6 Limitations

- Synthetic experiments are low-dimensional with G^* known by construction. This is a choice — total control of misalignment, impossible with human subjects — but generalization to high dimension remains to be demonstrated.
- Receivers are artificial. Human validation (experts vs. novices as different bases) is the next step, not the present contribution; no claim in this paper concerns human receivers.
- Misalignment is modeled by linear rotations (plus monotone deformations); real conceptual misalignments are nonlinear and structured.
- The “graph of M ” is a high-fidelity distilled proxy, not the exact network.
- Experiment 3’s dialogue/fine-tuning comparison is not information-equal (assumed — it is the thesis), and identifying the rule to edit is assisted by knowledge of G^* ; in real conditions, localizing the faulty rule is a problem in itself.
- The performance–interpretability trade-off [5]: our synthetic worlds are compressible by construction. The thesis “an interactive graph beats a frozen network under continual learning” is now tested in a synthetic regime (Experiment 7), not yet on real streams.
- Experiments 5–6: one dataset per verdict of the diptych; expert vocabularies are chosen by the authors, not learned; on AI4I, near-random residual failure modes limit the learnable structure of M .
- Experiment 7: sentences come from an oracle that localizes the concerned region (noisy variants tested; autonomous localization unresolved); the receiver has the world’s inductive bias — the $\theta = 1.5$ variant prices the opposite case.

7 Perspectives and conclusion

One line each, in order of priority: human validation of the framework (simulatability measured on subjects, bases operationalized by expertise); autonomous localization of the rule to edit when the correction does not name it; more datasets per verdict of the real-data diptych, with *learned* rather than chosen expert vocabularies; a formal proof of the monotone-invariance proposition and a characterization of the cost by mixing structure (angles between subspaces); contextual compression of large models — “the best graph of size K for context C ” — as a primitive; weighting explainability by realistic question distributions Q ; construction of a common basis through a bilateral feedback loop, superposition of crisp graphs as a model of vague language, and a calibration operator; and the connection to interactive proofs and debate [12], with the receiver as a bounded verifier.

Explainability research keeps asking what a model must look like to be understandable. We propose to ask, instead, *for whom* it is understandable — and to make the receiver a first-class citizen of the definition. Once the receiver is in the equation, explanation stops being a report and becomes a channel: measurable in questions (Experiments 1–2, 4),

predictive on real data in both directions (Experiments 5–6, including a clean pre-registered negative result), sensitive to the model’s own internal basis (Experiment 6), and reversible into a learning mechanism whose locality abolishes forgetting by construction (Experiments 3, 7). The evidence presented here is deliberately preliminary — controlled worlds, artificial receivers, honest kill conditions — but the direction is stated without hedging: there is no explainability without a receiver.

Reproducibility

All experiments (10 or 5 seeds, fixed), the `rex` library (rule-list model, W , W^{-1} , metrics, with unit tests including the edit-locality invariant), figures, and dataset download scripts are available at https://github.com/RedaAZIZI/impact_social. Experiment 7’s hypotheses and kill conditions were pre-registered in the repository before the run; Experiment 5’s protocol amendment (v0 $\rightarrow$ v0.1) is documented in the same place. No dataset is redistributed; both real datasets are public UCI benchmarks [21, 22].

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